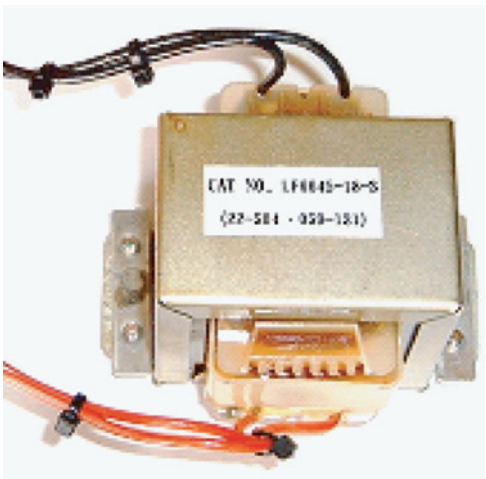
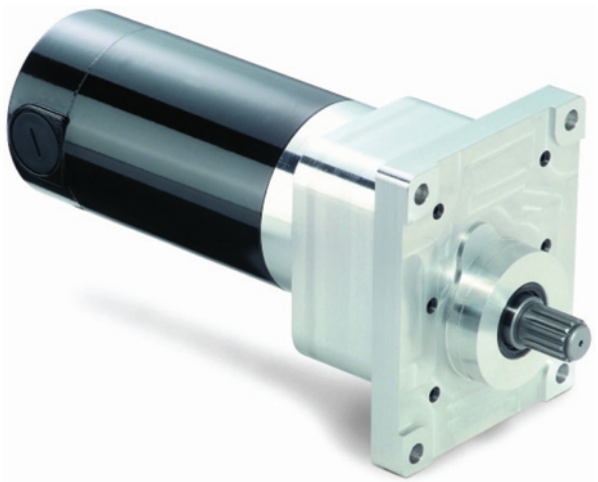
Chapter 15
Magnetic materials





Magnet materials in action. The audio-frequency transformer on the left has a soft magnetic core. The DC motor on the right has a hard (permanent) magnetic stator and a soft magnetic rotor

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15.1 Introduction and synopsis

Migrating birds, some think, navigate using the earth's magnetic field. This may be questionable, but what is beyond question is that sailors, for centuries, have navigated in this way, using a natural magnet, *lodestone*, to track it. Lodestone is a mineral, magnetite (Fe_3O_4), which sometimes occurs naturally in a magnetized state. Today we know lodestone as one of a family of *ferrites*, members of which can be found in every radio, television set and microwave oven. Ferrites are one of two families of magnetic material; the other is the *ferro-magnet* family, typified by iron but also including nickel, cobalt and alloys of all three. Placed in a magnetic field, these materials become magnetized, a phenomenon called *magnetic induction*; on removal, some, called soft magnets, lose their magnetization, others, the hard magnets, retain it.

Magnetic fields are created by moving electric charge—electric current in electromagnets, electron spin in atoms of magnetic materials. This chapter is about magnetic materials: how they are characterized, where their properties come from and how they are selected and used. It starts with definitions of magnetic properties and the way they are measured. As in other chapters, charts display them well, separating the materials that are good for one sort of application from those that are good for others. The chapter continues by drilling down to the origins of magnetic behavior, and concludes with a discussion of applications and the materials that best fill them.

15.2 Magnetic properties: definition and measurement

Magnetic fields in a vacuum

First, some definitions. When a current i passes through a long, empty coil of n turns and length L, as in Figure 15.1, a magnetic field is generated. The magnitude of the field, H, is given by Ampère's[1] law as

$$H = \frac{ni}{L} \tag{15.1}$$

and thus has units of amps/meter (A/m). The field has both magnitude and direction—it is a vector field.

Magnetic fields exert forces on a wire carrying an electric current. A current i flowing in a single loop of area S generates a dipole moment m, where

$$m = i\,S \tag{15.2}$$

[1] André Marie Ampère (1775–1836), French physicist, largely self-taught (and said by his fiancée to have 'no manners'), contributor to the theory of light, of electricity, of magnetism and to chemistry (he discovered fluorine). His father, a Justice of the Peace, was guillotined during the French Revolution.

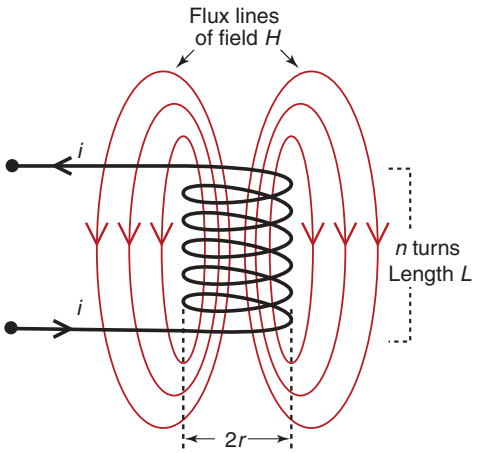


Figure 15.1 A solenoid creates a magnetic field *H*; the flux lines indicate the field strength.

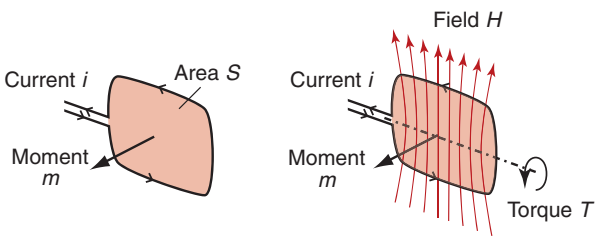


Figure 15.2 Definition of magnetic moment and moment-field interaction.

with units A.m^2, and it too is a vector with a direction normal to the plane of S (Figure 15.2). If the loop is placed at right angles to the field H it feels a torque T (units: newton.meter, or N.m) of

$$T = \mu_o \, m \, H \tag{15.3}$$

where μ_o is called the *permeability of a vacuum*, $\mu_o = 4\pi \times 10^{-7}$ henry/meter (H/m). To link these we define a second measure of the magnetic field, one that relates directly to the torque it exerts on a unit magnetic moment. It is called the *magnetic induction* or *flux density*, B, and for a vacuum or non-magnetic materials it is

$$B = \mu_o \, H \tag{15.4}$$

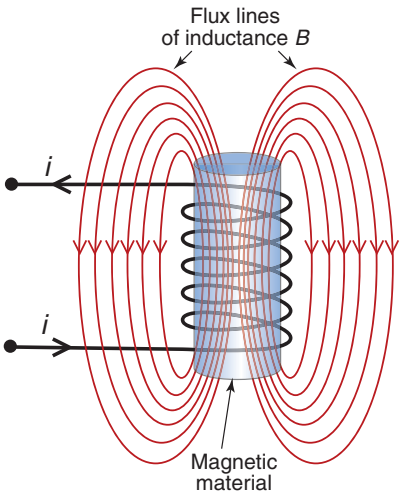


Figure 15.3 A magnetic material exposed to a field H becomes magnetized, concentrating the flux lines.

Its units are *tesla*[2], so 1 tesla is 1 HA/m^2. A magnetic induction B of 1 tesla exerts a torque of 1 N.m on a unit dipole at right angles to the field H.

Magnetic fields in materials

If the space inside the coil of Figure 15.2 is filled with a material, as in Figure 15.3, the induction within it changes. This is because its atoms respond to the field by forming little magnetic dipoles in ways that are explained in Section 15.4. The material acquires a macroscopic dipole moment or *magnetization*, M (its units are A/m, like H). The induction becomes

$$B = \mu_o (H + M) \tag{15.5}$$

The simplicity of this equation is misleading, since it suggests that M and H are independent; in reality M is the response of the material to H, so the two are coupled. If the material of the core is ferro-magnetic, the response is a very strong one and it is nonlinear, as we shall see in a moment. It is usual to rewrite equation (15.5) in the form

$$B = \mu_R \mu_o H$$

where μ_R is called the *relative permeability*, and like the relative permittivity (the dielectric constant) of Chapter 14, it is dimensionless. The magnetization, M, is thus

$$M = (\mu_R - 1) H = \chi H \tag{15.6}$$

[2] Nikola Tesla (1856–1943), Serbian-American inventor, discoverer of rotating magnetic fields, the basis of most alternating current machinery, inventor of the Tesla coil and of a rudimentary flying machine (never built).

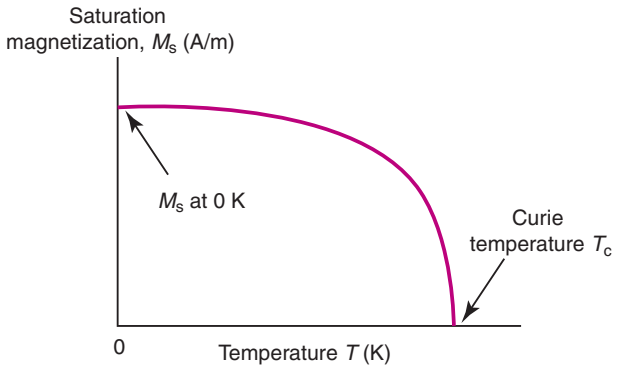


Figure 15.4 Saturation magnetization decreases with temperature, falling to zero at the Curie temperature, T_c.

where χ is the *magnetic susceptibility*. Neither μ_R nor χ are constants—they depend not only on the material but also on the magnitude of the field, H, for the reason just given.

Nearly all materials respond to a magnetic field by becoming magnetized, but most are paramagnetic with a response so faint that it is of no practical use. A few, however, contain atoms that have large dipole moments and have the ability to spontaneously magnetize—to align their dipoles in parallel—as electric dipoles do in ferro-electric materials. These are called *ferro-magnetic* and *ferri-magnetic* materials (the second one is called *ferrites* for short), and it is these that are of real practical use.

Magnetization decreases with increasing temperature. Just as with ferro-electrics, there is a temperature, the Curie temperature T_c, above which it disappears, as in Figure 15.4. Its value for the materials we shall meet here is well above room temperature (typically 300–500°C) but making magnets for use at really high temperatures is a problem.

Measuring magnetic properties

Magnetic properties are measured by plotting an M–H curve. It looks like Figure 15.5. If an increasing field H is applied to a previously demagnetized sample, starting at A on the figure, its magnetization increases, slowly at first and then faster, following the broken line, until it finally tails off to a maximum, the *saturation magnetization* M_s at the point B. If the field is now backed off, M does not retrace its original path, but retains some of its magnetization so that when H has reached zero, at the point C, some magnetization remains: it is called the *remanent magnetization* or *remanence* M_R and is usually only a little less than M_s. To decrease M further we must increase the field in the opposite direction until M finally passes through zero at the point D when the field is $-H_c$, the *coercive field*, a measure of the resistance to demagnetization. Some applications require H_c to be as high as possible, others as low a possible. Beyond

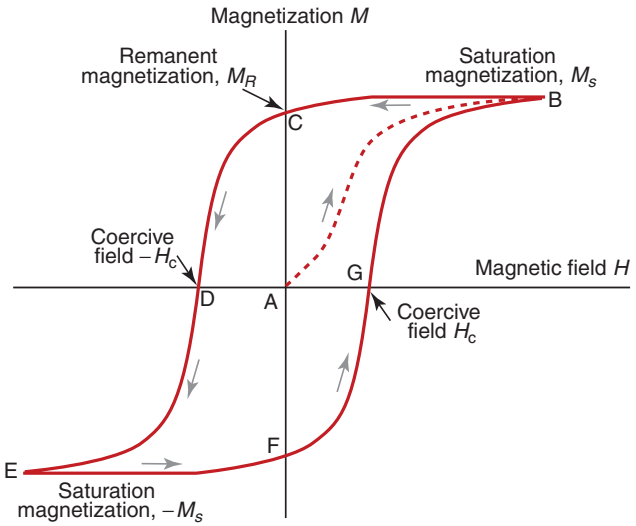


Figure 15.5 A hysteresis curve, showing the important magnetic properties.

point D the magnetization M starts to increase in the opposite direction, eventually reaching saturation again at the point E. If the field is now decreased again, M follows the curve through F and G back to full forward magnetic saturation again at B to form a closed M–H circuit called the *hysteresis loop*.

Magnetic materials are characterized by the size and shape of their hysteresis loops. The initial segment AB is called the *initial magnetization curve* and its average slope (or sometimes its steepest slope) is the magnetic susceptibility, χ. The other key properties—the saturation magnetization M_s, the remanence M_R and the coercive field H_c—have already been defined. Each full cycle of the hysteresis loop dissipates an energy per unit volume equal to the area of the loop multiplied by μ_o, the permeability of a vacuum. This energy appears as heat (it is like magnetic friction). Many texts plot not the M–H curve, but the curve of inductance B against H. Equation (15.5) says that B is proportional to $(M + H)$, and the value of M for any magnetic materials worthy of the name is very much larger than the H applied to it, so $B \approx \mu_o M$ and the B–H curve of a ferro-magnetic material looks very like its M–H curve (it's just that the M-axis has been scaled by μ_o).

There are several ways to measure the hysteresis curve, one of which is sketched in Figure 15.6. Here the material forms the core of what is, in effect, a transformer. The oscillating current through the primary input coil creates a field H that induces magnetization M in the material of the core, driving it round its hysteresis loop. The secondary coil picks up the inductance, from which the instantaneous state of the magnetization can be calculated, mapping out the loop.

Magnetic materials differ greatly in the shape and area of their hysteresis loop, the greatest difference being that between *soft magnets*, which have thin

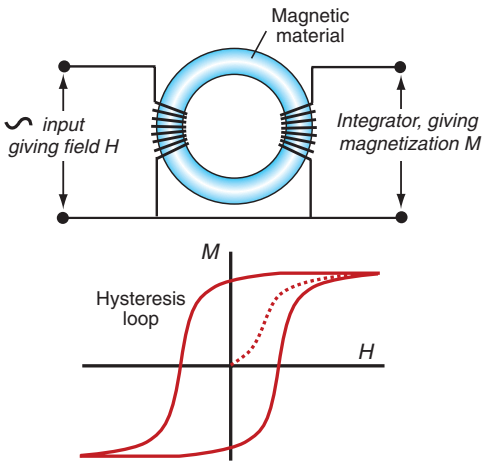


Figure 15.6 Measuring the hysteresis curve.

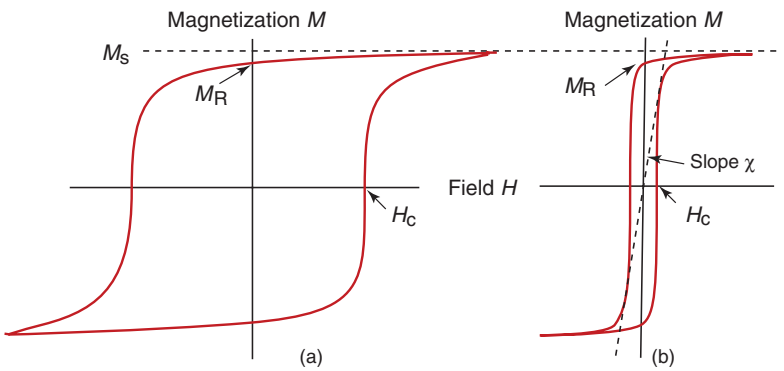


Figure 15.7 Hysteresis loops. (a) A fat loop typical of hard magnets. (b) A thin, square loop typical of soft magnets.

loops, and *hard magnets*, which have fat ones, as sketched in Figure 15.7. In fact the differences are much greater than this figure suggests: the coercive field H_c (which determines the width of the loop) of hard magnetic materials like Alnico is greater by a factor of about 10^5 than that of soft magnetic materials like silicon–iron. More on this in the next two sections.

15.3 Charts for magnetic properties

The remanence–coercive field chart

The differences between families of soft and hard magnetic materials are brought out by Figure 15.8. The axes are remanent magnetization M_r and coercive field

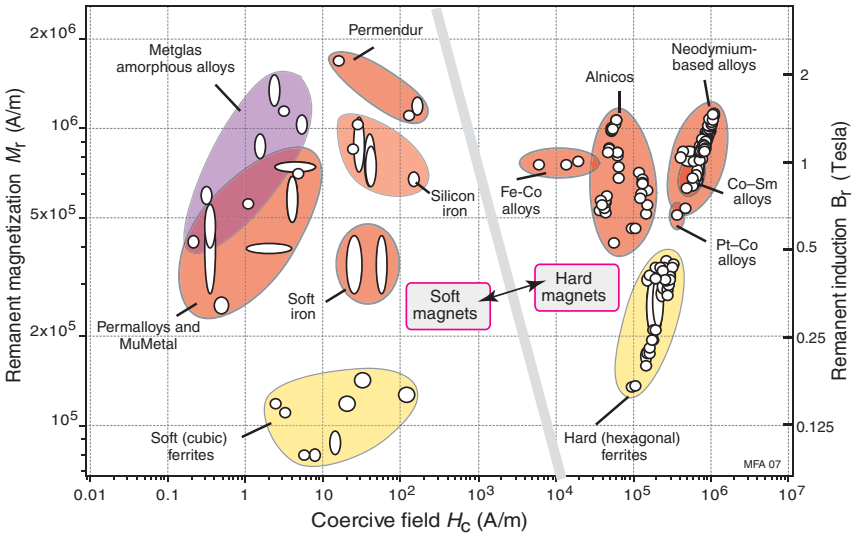


Figure 15.8 Remanent magnetization and coercive force. Soft magnetic materials lie on the left, hard magnetic materials on the right. The red and purple envelopes enclose electrically conducting materials, the yellow ones enclose insulators.

H_c. The saturation magnetization M_s, more relevant for soft magnets, is only slightly larger than M_r, so an M_s–H_c chart looks almost the same as this one. There are 12 district families, each enclosed in a colored envelope. The members of each family, shown as smaller ellipses, have unhelpful tradenames (such as 'H FerriteYBM-1A') so they are not individually labeled.[3] Soft magnets require high M_s and low H_c; they are the ones on the left, with the best near the top. Hard magnets must hold their magnetism, requiring a high H_c, and to be powerful they need a large M_r; they are the ones on the right, with the best again at the top.

In many applications the electrical resistivity is also important because of eddy-current losses, described later. Changing magnetic fields induces eddy currents in conductors but not in insulators. The red and purple ellipses on the chart enclose metallic ferro-magnetic materials, which are good conductors. The yellow ones describe ferrites, which are insulators.

The saturation magnetization–susceptibility chart

This is the chart for selecting soft magnetic materials (Figure 15.9). It shows the saturation magnetization, M_s—the ultimate degree to which a material can concentrate magnetic flux—plotted against its susceptibility, χ, which measures the ease with which it can be magnetized. Many texts use, instead, the saturation inductance B_s and maximum relative permeability, μ_R. They are shown on the

[3] This chart and the next were made with Level 3 of the CES Edu database, which contains data for 222 magnetic materials. Levels 1 and 2 do not include magnetic materials because of their specialized nature.

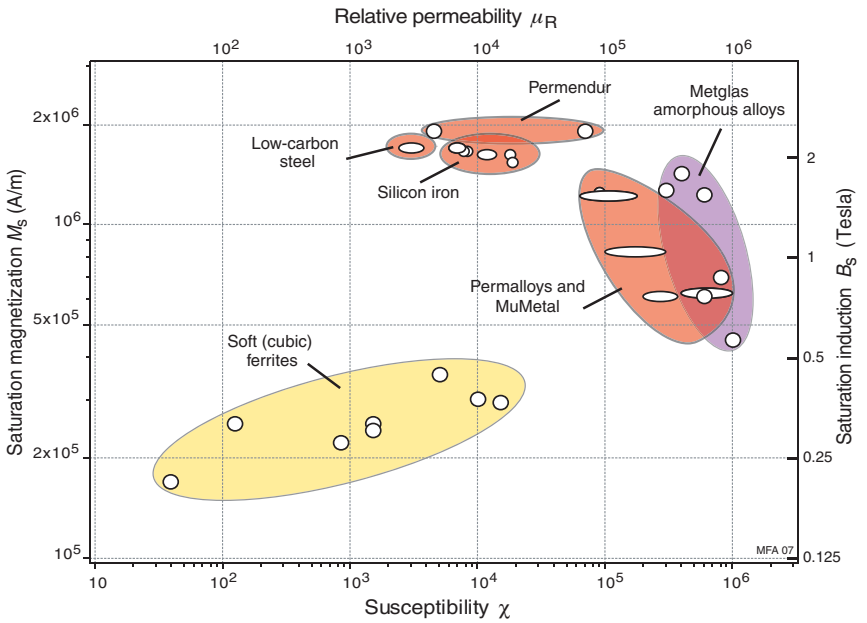


Figure 15.9 Saturation magnetization and susceptibility for soft magnetic materials. The red and purple envelopes enclose electrically conducting materials, the yellow ones enclose insulators.

other axes of the chart. As in the first chart, materials that are electrical conductors are enclosed in red or purple ellipses, those that are insulators yellow ones.

15.4 Drilling down: the physics and manipulation of magnetic properties

The classical picture of an atom is that of a nucleus around which swing electrons, as in Figure 15.10. Moving charge implies an electric current, and an electric current flowing in a loop creates a magnetic dipole, as in Figure 15.2. There is, therefore, a magnetic dipole associated with each orbiting electron. That is not all. Each electron has an additional moment of its own: its spin moment. A proper explanation of this requires quantum mechanics, but a way of envisaging its origin is to think of an electron not as a point charge but as slightly spread out and spinning on its own axis, again creating rotating charge and a dipole moment—and this turns out to be large. The total moment of the atom is the vector sum of the whole lot.

A simple atom like that of helium has two electrons per orbit and they configure themselves such that the moment of one exactly cancels the moment of

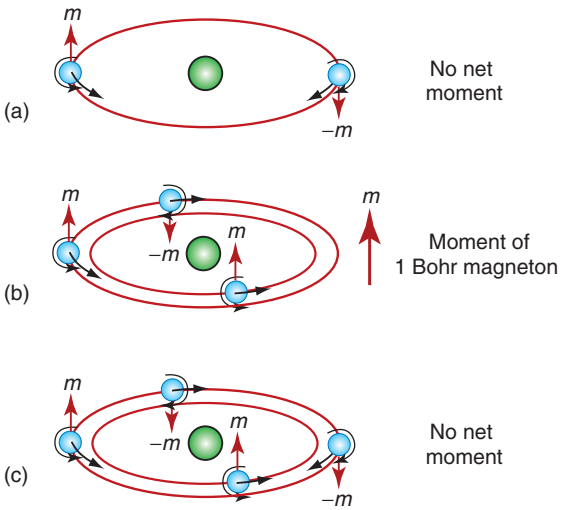


Figure 15.10 Orbital and electron spins create a magnetic dipole. Even numbers of electrons filling energy levels in pairs have moments that cancel, as in (a) and (c). An unpaired electron gives the atom a permanent magnetic moment, as in (b).

the other, as in Figure 15.10(a) and (c), leaving no net moment. But now think of an atom with three, not two, electrons as in (b). The moments of two may cancel, but there remains the third, leaving the atom with a net moment represented by the red arrow at the right of Figure 15.10(b). Thus, atoms with electron moments that cancel are non-magnetic; those with electron moments that don't cancel carry a magnetic dipole. Simplifying a little, one unpaired electron gives a magnetic moment of 9.3×10^{-24} A.m^2, called a Bohr[4] magneton; two unpaired electrons give two Bohr magnetons, three gives three and so on.

Think now of the magnetic atoms assembled into a crystal. In most materials the atomic moments interact so weakly that thermal motion is enough to randomize their directions, as in Figure 15.11(a). Despite their magnetic atoms, the structure as a whole has no magnetic moment; these materials are *paramagnetic*. In a few materials, though, something quite different happens. The fields of neighboring atoms interact such that their energy is reduced if their magnetic moments line up. This drop in energy is called the *exchange energy* and it is strong enough that it beats the randomizing effect of thermal energy so long as the temperature is not too high (the shape of the Curie curve of Figure 15.4 shows how thermal energy overwhelms the exchange energy as the Curie temperature is approached). They may line up anti-parallel, head to tail so to speak, as in Figure 15.11(b), and there is still no net moment; such materials are called *anti-ferro-magnets*.

[4] Niels Henrik David Bohr (1885–1962), Danish theoretical physicist, elucidator of the structure of the atom, contributor to the Manhattan Project and campaigner for peace.

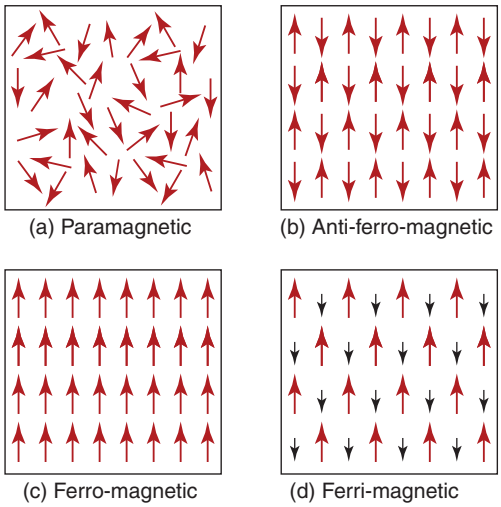


Figure 15.11 Types of magnetic behavior.

But in a few elements, notably iron, nickel and cobalt, exactly the opposite happens: the moments spontaneously align so that—if all are parallel—the structure has a net moment that is the sum of those of all the atoms it contains. These materials are *ferro-magnetic*. Iron has three unpaired electrons per atom, nickel has two and cobalt just one, so the net moment if all the spins are aligned (the saturation magnetization M_s) is greatest for iron, less for nickel and still less for cobalt.

Compounds give a fourth possibility. The materials we have referred to as ferrites are oxides; one class of them has the formula MFe_2O_4, where M is also a magnetic atom, such as cobalt (Co). Both the iron and the cobalt atoms have dipoles but they differ in strength. They line up in the anti-parallel configuration, but because of the difference in moments the cancelation is incomplete, leaving a net moment M; these are *ferri-magnets*, ferrites for short. The partial cancelation and the smaller number of magnetic atoms per unit volume means they have lower saturation magnetization than, say, iron, but they have other advantages, notably that, being oxides, they are electrical insulators.

Domains

If atomic moments line up, shouldn't every piece of iron, nickel or cobalt be a permanent magnet? Magnetic materials they are; magnets, in general, they are not. Why not?

A uniformly magnetized rod creates a magnetic field, H, like that of a solenoid. The field has a potential energy

$$U = \frac{1}{2}\mu_o \int_V H^2 dV \tag{15.7}$$

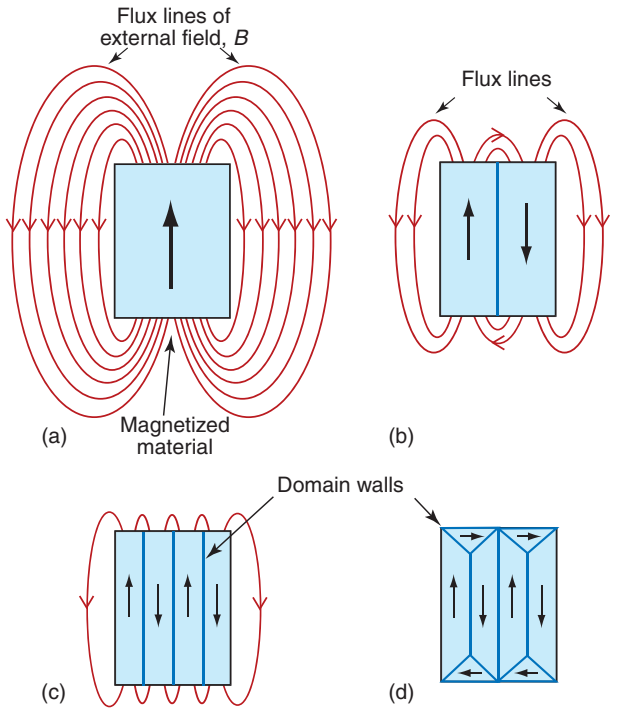


Figure 15.12 Domains allow a compromise: the cancelation of the external field while retaining magnetization of the material itself. The arrows show the direction of magnetization.

where the integral is carried out over the volume V within which the field exists. Working out this integral is not simple, but we don't need to do it. All we need to note is that the smaller is H and the smaller the volume V that it invades, the smaller is the energy. If the structure can arrange its moments to minimize its H or get rid of it entirely (remembering that the exchange energy wants neighboring atom moments to stay parallel) it will try to do so.

Figure 15.12 illustrates how this can be done. The material splits up into *domains*, within which the atomic moments are parallel but with a switch of direction between mating domains to minimize the external field. The domains meet at *domain walls*, regions a few atoms thick in which the moments swing from the orientation of one domain to that of the other. Splitting into parallel domains of opposite magnetization, as in (b) and (c), reduces the field a lot; adding caps magnetized perpendicular to both as in (d) kills it almost completely. The result is that most magnetic materials, unless manipulated in some way, adopt a domain structure with minimal external field—that is the same as saying that, while magnetic, they are not magnets.

How can they be magnetized? Placed in a magnetic field created, say, with a solenoid, the domains already aligned with the field have lower energy than those